

Problem Set 4

1. Difference equation to $H(z)$

A system obeys $y[n] - 1.4y[n-1] + 0.45y[n-2] = x[n] + x[n-1]$.

- Find $H(z)$.
- Determine **all** zeros and poles of $H(z)$.
- Determine the ROC if the system is causal.
- Is the causal system BIBO stable?

2. Geometric frequency-response reasoning

Consider $H(z) = (1 + z^{-1})/(1 - 0.8z^{-1})$.

- Determine the poles and zeros of $H(z)$.
- Use a geometric method to compare $|H(e^{j\omega})|$ near $\omega = 0$ and $\omega = \pi$. Show a labeled sketch of the geometric evaluation.
- What is your conclusion about the qualitative filter type of this $H(z)$?

3. 600 Hz notch filter design

At $f_s = 9000$ Hz, design the pole-zero locations of a second-order notch at 600 Hz using zeros on the unit circle and poles at radius $r = 0.96$ with the same angles ω_0 .

- Give ω_0 , the two zero locations and the two pole locations.
- Explain what increasing r to 0.99 would do.

4. Ideal lowpass FIR filter and causal realization

An ideal discrete-time lowpass filter has

$$H_d(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \omega_c, \\ 0, & \omega_c < |\omega| \leq \pi. \end{cases}$$

- Use the inverse DTFT to determine the unit impulse response $h_d[n]$
- Sketch and label $h_d[n]$ for $\omega_c = \pi/3$ and n in the range $[-8, 8]$. What is the value of $h_d[0]$?
- Explain why this ideal impulse response cannot be implemented exactly as a real-time FIR filter.
- For an odd-length filter with $N = 61$ taps, state the shift required to make a symmetric truncated version causal and give its constant group delay.

5. Window tradeoff

A student designs two 51-tap lowpass filters with the same cutoff. Filter A uses a rectangular window; Filter B uses a Hamming window.

- a) Which filter should have the narrower transition band?
- b) Which filter should have lower stopband sidelobes?
- c) Give approximate first-sidelobe levels for the two windows.
- d) Explain why merely increasing the rectangular-window length does not remove the Gibbs overshoot near a discontinuity.

6. Symmetry and linear phase

Let a real-valued FIR filter have length N and satisfy

$$h[n] = h[N - 1 - n].$$

- a) Show that its frequency response can be written in the form

$$H(e^{j\omega}) = e^{-j(N-1)\omega/2} A(\omega),$$

where $A(\omega)$ is real-valued.

- b) What is the group delay where $A(\omega) \neq 0$? Why can this be useful in machine learning in a feature-extraction pipeline?

7. Pole radius and time-domain decay (ECEN 5632 Problem)

A causal system has a complex pole pair $p = r e^{\pm j\theta}$. Explain why the corresponding impulse-response terms decay roughly as r^n and why moving r toward 1 simultaneously increases ringing duration and sharpens the frequency-domain resonance. Connect your argument to geometric distance from the unit circle.